Discovery Lab

Exam 5 Revision

Lower Secondary Stage 2 · Science · Lessons 23 to 28

Breathing and respiration, bones and muscles, medicines, states of matter, diffusion and air.



- Lesson 23: Breathing and Respiration
- Lesson 24: Bodies and Bones
- Lesson 25: Moving Bones and Medicines
- Lesson 26: Summary of Properties
- Lesson 27: Measuring Movements
- Lesson 28: Air and Particles

Name: _____ Class: _____ Date: _____

LESSON 23

23 Breathing and Respiration

Two different processes that work together.

Key ideas

Breathing (ventilation) moves air in and out of the lungs. It brings in **oxygen** and removes **carbon dioxide**. It is a **physical** process.

Respiration is a **chemical reaction** in cells: **glucose + oxygen → carbon dioxide + water + energy**. It happens in the **mitochondria**.

Air path: nose or mouth → trachea → bronchi → lungs → **alveoli** (gas exchange).

Athletes breathe faster to get more oxygen for respiration.

Feature	Breathing	Respiration
Where	Lungs	All body cells
What it moves	Air (oxygen in, CO_2 out)	Oxygen and glucose → energy
Type	Physical process	Chemical reaction
Needed for	Gas exchange	Releasing energy

DID YOU KNOW?

Your lungs take in about 11,000 litres of air every day.

Check yourself

- [ ] What is breathing?
- [ ] What is respiration?
- [ ] Where does respiration happen?
- [ ] Why do athletes breathe faster?
- [ ] What is the job of the alveoli?

REVISION TASK

Draw the path of air from the nose to the alveoli and label each part.

LESSON 24

24 Bodies and Bones

The skeleton: support, protection and movement.

Key ideas

An adult skeleton has **206 bones**. Jobs: **support**, **protection**, **movement**.

Ligaments join bone to bone and stabilise joints. **Tendons** join muscle to bone. **Cartilage** cushions joints.

Bone marrow makes red blood cells. Bones are hard (**calcium**) but not brittle (**collagen**).

Babies have more bones than adults because some bones **fuse** as they grow.

Bone	Job
Skull	Protects the brain
Ribcage	Protects the lungs and heart
Spine	Supports the body and protects the spinal cord
Pelvis	Connects the legs to the hips
Femur	Longest bone; supports the body

DID YOU KNOW?

The smallest bone is the stapes in your ear, about the size of a grain of rice.

Check yourself

- [ ] What are the three jobs of the skeleton?
- [ ] What connects bone to bone?
- [ ] What does bone marrow do?
- [ ] Why are bones hard but not brittle?
- [ ] Why do babies have more bones?

REVISION TASK

Draw a skeleton and label the skull, ribcage, spine, pelvis and femur with their jobs.

LESSON 25

25 Moving Bones and Medicines

How muscles move bones, and how common medicines help.

Key ideas

Muscles **contract** and pull on bones through **tendons**. They work in **antagonistic pairs**: one contracts while the other relaxes.

Biceps bends the arm; **triceps** straightens it. **Quadriceps** straightens the knee; **hamstrings** work at the back of the thigh. **Deltoid** raises the arm.

Medicines: **paracetamol** lowers fever and pain; **antihistamine** blocks histamine; **decongestant** opens blocked airways; **antibiotics** kill bacteria, **antivirals** kill viruses. Antibiotics do not work on colds.

Safe use: read the label, only from a trusted adult or doctor, never share prescriptions.

Medicine	What it does
Antibiotic	Kills bacteria
Paracetamol	Reduces fever and pain
Decongestant	Opens blocked airways
Antihistamine	Blocks histamine
Antiviral	Kills viruses

DID YOU KNOW?

You have over 600 muscles. The strongest for its size is the jaw muscle (masseter).

Check yourself

- [ ] What happens when a muscle contracts?
- [ ] Why do muscles work in pairs?
- [ ] What connects muscle to bone?
- [ ] Why don't antibiotics help with colds?
- [ ] What should you do before taking medicine?

REVISION TASK

Act out the robot arm, then draw the biceps and triceps when the arm is bent and straight.

LESSON 26

26 Summary of Properties

Solids, liquids and gases, and how particles explain them.

Key ideas

Solid: particles tightly packed, vibrate in place, cannot be compressed. **Liquid**: close, slide over each other, cannot be compressed. **Gas**: far apart, fast in all directions, can be compressed.

A solid keeps its shape and volume; a liquid changes shape but not volume; a gas fills its container.

Changes of state: **melting** (solid → liquid), **freezing** (liquid → solid), **evaporation** (liquid → gas), **condensation** (gas → liquid), **sublimation** (solid → gas).

All are **physical changes**: the particles are rearranged, not changed chemically.

State	Arrangement	Movement	Compressed?
Solid	Tightly packed	Vibrate in a fixed place	No
Liquid	Close, can move past	Slide over each other	No
Gas	Far apart	Fast in all directions	Yes

DID YOU KNOW?

Clouds form when water vapour condenses into tiny liquid droplets.

Check yourself

- [ ] Which state can be compressed easily?
- [ ] What is melting?
- [ ] What is sublimation?
- [ ] Why can gases be compressed but solids cannot?
- [ ] Which state fills its container completely?

REVISION TASK

Draw particle diagrams for a solid, a liquid and a gas and label the spacing and movement.

LESSON 27

27 Measuring Movements

Particles never stop moving: Brownian motion and diffusion.

Key ideas

Brownian motion: the random movement of particles in liquids and gases because other particles hit them. First seen in **pollen grains** in water.

Diffusion: particles spread from a **high** to a **low** concentration. It happens in liquids and gases, not solids.

Examples: perfume in a room, ink in water, oxygen moving into your blood.

Higher temperature = faster particles = **faster diffusion**.

Idea	Meaning
Brownian motion	Random movement caused by particle collisions
Diffusion	Spreading from high to low concentration
Higher temperature	Faster movement and diffusion
Lower temperature	Slower movement and diffusion

DID YOU KNOW?

You can smell popcorn across a room because gas particles diffuse.

Check yourself

- [ ] What causes Brownian motion?
- [ ] What is diffusion?
- [ ] In which states does diffusion happen?
- [ ] What happens to diffusion when it gets hotter?
- [ ] Why does perfume spread across a room?

REVISION TASK

Draw ink spreading in water at the start, after one minute and after ten minutes.

LESSON 28

28 Air and Particles

What air is made of, and why it pushes on everything.

Key ideas

Air is a **mixture** of gases: about **78% nitrogen**, **21% oxygen** and **1%** others (carbon dioxide, argon...).

Air particles move fast in all directions, have mass and take up space.

When particles **collide** with surfaces they make **air pressure**. More particles or faster particles = higher pressure.

Oxygen is for breathing and respiration; carbon dioxide is used by plants; argon is inert; water vapour helps control temperature and humidity.

Gas	Job
Oxygen	Needed for breathing and respiration
Nitrogen	Makes up most of the air
Carbon dioxide	Used by plants in photosynthesis
Argon	Inert gas used in balloons and lights
Water vapour	Helps control temperature and humidity

DID YOU KNOW?

Air pressure can crush a metal can if the air inside is removed.

Check yourself

- [ ] What percentage of air is nitrogen?
- [ ] What percentage is oxygen?
- [ ] What causes air pressure?
- [ ] What happens if you remove air from a can?
- [ ] What happens to pressure when particles move faster?

REVISION TASK

Make a pie chart of the gases in air and label each gas with its job.

FINAL PRACTICE

Final practice

Tick each box when you can answer without looking.

Final practice

- [ ] Compare breathing and respiration, including where each happens and its function.
- [ ] Explain how bones and muscles work together to produce movement.
- [ ] Describe the particle arrangement, movement and energy in solids, liquids and gases.
- [ ] Explain why perfume smells spread faster in a warm room.
- [ ] Why should antibiotics only be taken when prescribed?

CHALLENGE

Link two topics: explain how breathing, respiration and muscles are connected when you run.

Revision checklist

Revision task	Done?
Reviewed the Canvas slides for Lessons 23–28	☐
Answered every “Check yourself” question	☐
Practised the matching and table activities	☐
Used the 3D labs: breathing, muscles and particles	☐
Practised short answers and the practice exam	☐
Reviewed the keywords: respiration, tendon, diffusion, pressure	☐

Try it live · Practise with 3D labs, games and a practice exam at **morehumaned.com/discovery-lab**

ANSWER KEY

Answer key

Check your own answers against these. If one is hard, go back to the lesson.

Lesson 23: Breathing and Respiration

1. Moving air in and out of the lungs.
2. A chemical reaction in cells that releases energy.
3. In the mitochondria of cells.
4. Their cells need more oxygen to release more energy.
5. Gas exchange between the air and the blood.

Lesson 24: Bodies and Bones

1. Support, protection and movement.
2. Ligaments.
3. It makes red blood cells.
4. Calcium gives strength and collagen gives flexibility.
5. Some bones fuse as they grow.

Lesson 25: Moving Bones and Medicines

1. It shortens and pulls on the bone.
2. One pulls the bone one way and the other pulls it back.
3. Tendons.
4. Colds are caused by viruses, not bacteria.
5. Read the label and check with a trusted adult.

Lesson 26: Summary of Properties

1. Gas.
2. A solid turning into a liquid.
3. A solid changing directly into a gas.
4. Gas particles are far apart and can be pushed closer.
5. Gas.

Lesson 27: Measuring Movements

1. Particles being hit by other moving particles.
2. Particles spreading from high to low concentration.
3. Liquids and gases.
4. It gets faster.
5. Its particles move randomly and spread out.

Lesson 28: Air and Particles

1. About 78%.
2. About 21%.
3. Air particles colliding with surfaces.
4. The air outside crushes the can.
5. The pressure increases.

Final practice

1. Breathing moves air in the lungs; respiration releases energy in all cells.
2. Muscles contract and pull on bones through tendons; they work in pairs.
3. Solids: tightly packed, vibrating; liquids: close, sliding; gases: far apart, fast; energy increases from solid to gas.
4. Warm particles move faster, so diffusion is faster.
5. They only kill bacteria and misuse makes bacteria resistant.

Discovery Lab

You are going to do great! Use this booklet to plan your study time, then bring it all together.

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